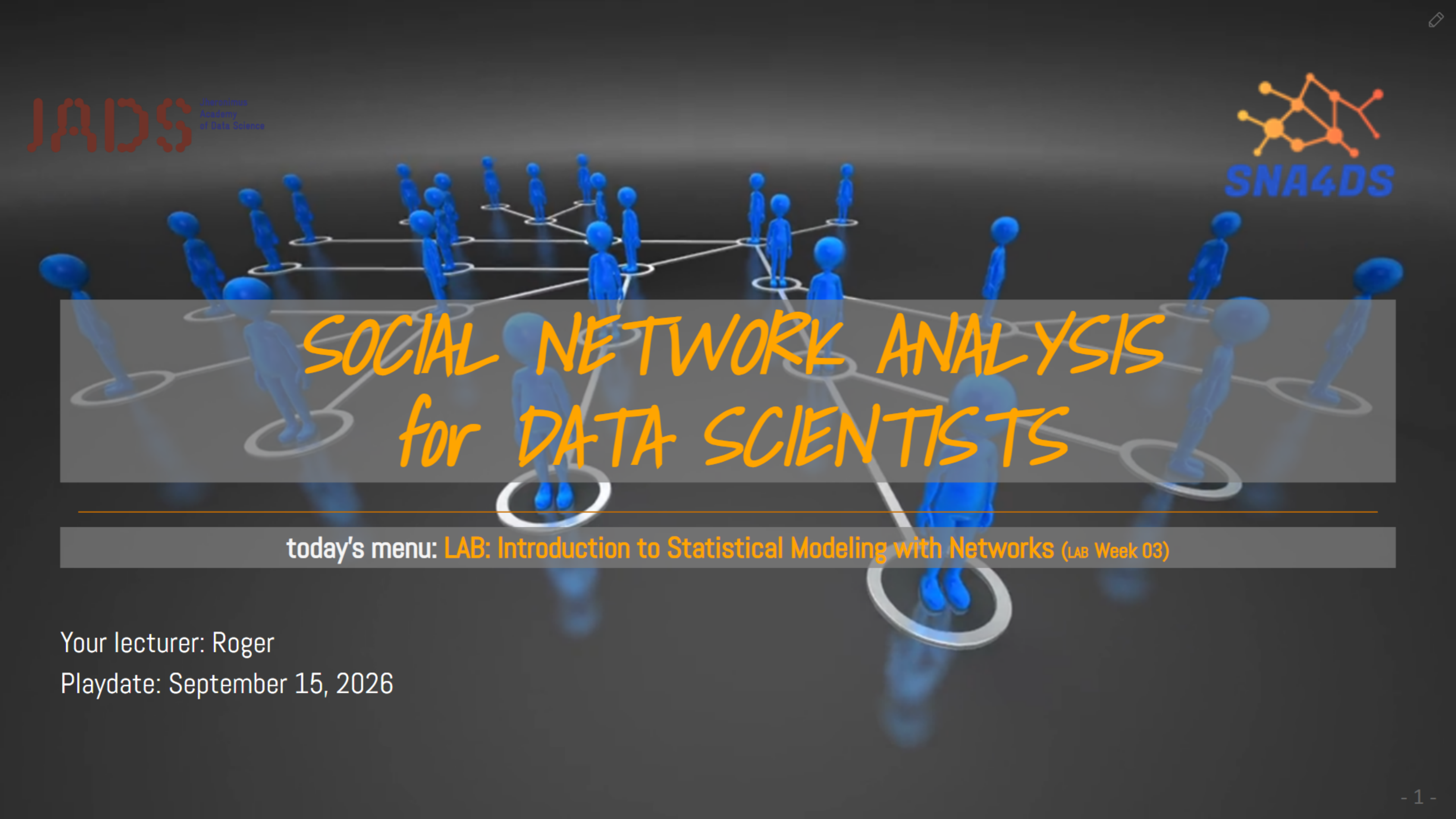




SOCIAL NETWORK ANALYSIS for DATA SCIENTISTS

today's menu: LAB: Introduction to Statistical Modeling with Networks (LAB Week 03)

Your lecturer: Roger

Playdate: September 15, 2026

Today's menu

1. Homeplay
2. NAM findings
3. CUG findings and extension
4. Assessment and practical overview of the methods

Today's menu



Homeplay

How did the tutorial and homeplay go?

All clear?

Challenges?

useful function to extract vertex attributes



```
attrs <- snafun::extract_all_vertex_attributes(louis)
attrs
```

	black_political_equality	democratic	Intercept	perc_black	perc_cath	perc_urban	vertex.names
1	80.7	75.8	1	19.7	83.0	55.8	1
2	79.0	59.8	1	24.8	45.4	33.3	2
3	68.6	74.8	1	31.9	88.2	33.4	3
4	68.2	71.7	1	41.2	95.9	0.0	4
5	42.8	76.0	1	27.8	87.1	25.1	5
6	55.3	48.7	1	22.4	11.0	37.5	6
7	1.0	17.6	1	49.4	0.2	15.2	7
8	16.5	25.2	1	24.9	9.0	66.0	8
9	20.5	24.8	1	36.5	10.2	80.8	9
10	69.4	64.4	1	20.9	52.6	73.9	10
11	7.9	34.0	1	27.8	1.3	0.0	11
12	75.0	83.1	1	6.4	78.8	0.0	12
13	23.9	26.2	1	35.2	0.0	0.0	13
14	1.0	12.7	1	50.3	1.5	39.7	14
15	14.5	23.1	1	46.3	10.4	43.4	15
16	16.2	26.7	1	57.5	10.4	24.1	16
17	43.4	46.6	1	31.8	36.6	85.1	17
18	0.0	23.9	1	61.2	27.7	40.1	18

R code NAM

NAM Output: lag model

```
mod <- snafun::stat_nam(democratic ~ perc_black +  
                        perc_cath + perc_urban + black_political_equality,  
                        data = attrs, W = louis)
```

```
Call:snafun::stat_nam(formula = democratic ~ perc_black + perc_cath +
  perc_urban + black_political_equality, data = attrs, W = louis)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.68996	-4.80434	0.58857	4.05038	17.69156

Type: lag

Coefficients: (asymptotic standard errors)

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	13.8735674	4.6729365	2.9689	0.002988
perc_black	-0.0040543	0.0718224	-0.0564	0.954984
perc_cath	0.2255374	0.0464735	4.8530	1.216e-06
perc_urban	-0.0979634	0.0365971	-2.6768	0.007433
black_political_equality	0.2957095	0.0589806	5.0137	5.340e-07

Rho: 0.30601, LR test value: 7.8801, p-value: 0.0049982

Asymptotic standard error: 0.096258

z-value: 3.179, p-value: 0.0014778

Wald statistic: 10.106, p-value: 0.0014778

Log likelihood: -216.5846 for lag model

ML residual variance (sigma squared): 49.81, (sigma: 7.0576)

Number of observations: 64

R code NAM

NAM Output: combined model

```
mod2 <- snafun::stat_nam(democratic ~ perc_black +  
  perc_cath + perc_urban + black_political_equality,  
  data = attrs, W = louis, W2 = louis, model = "combined")
```

```
Call:snafun::stat_nam(formula = democratic ~ perc_black + perc_cath +
  perc_urban + black_political_equality, data = attrs, W = louis,
  W2 = louis, model = "combined")
```

Residuals:

Min	1Q	Median	3Q	Max
-18.41300	-3.59401	0.12036	3.43756	19.56223

Type: sac

Coefficients: (asymptotic standard errors)

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	21.541054	6.267568	3.4369	0.0005884
perc_black	-0.096849	0.072257	-1.3403	0.1801373
perc_cath	0.335143	0.053757	6.2343	4.537e-10
perc_urban	-0.074861	0.030946	-2.4191	0.0155580
black_political_equality	0.243187	0.055839	4.3551	1.330e-05

Rho: 0.14193

Asymptotic standard error: 0.1263

z-value: 1.1238, p-value: 0.26111

Lambda: 0.61507

Asymptotic standard error: 0.13887

z-value: 4.4291, p-value: 9.4639e-06

FIFA 2006 and 2015 networks

What were your findings for these networks?

Transitivity: conditioning on edges

R code

Output

Output plot

```
trans_f <- function(x, directed = FALSE) {  
  x <- snafun::fix_cug_input(x, directed = directed)  
  snafun::g_transitivity(x)  
}  
  
cug_trans_15 <- sna::cug.test(fifa2015, mode = "graph",  
                             FUN = trans_f,  
                             cmode = "edges", reps = 1000)  
  
cug_trans_15
```

OR

```
cug_trans_15 <- snafun::stat_cug(fifa2015, mode = "graph",  
                                FUN = snafun::g_transitivity,  
                                cmode = "edges", reps = 1000)  
  
cug_trans_15
```

Transitivity: conditioning on edges



R code

Output

Output plot

Univariate Conditional Uniform Graph Test

Conditioning Method: edges

Graph Type: digraph

Diagonal Used: FALSE

Replications: 200

Observed Value: 0.6529858

$\Pr(X \geq \text{Obs})$: 0

$\Pr(X \leq \text{Obs})$: 1

Transitivity: conditioning on edges

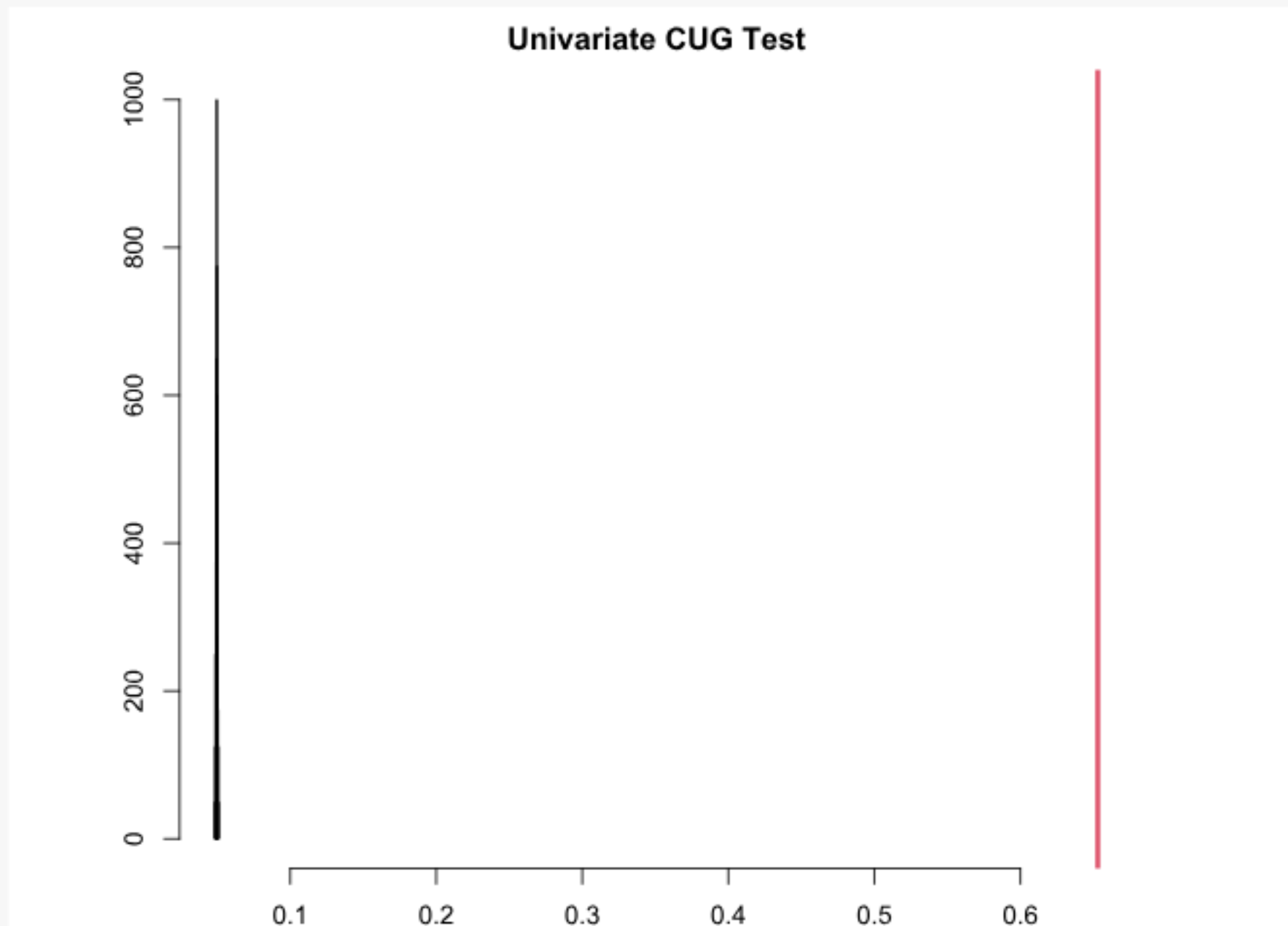
R code

Output

Output plot

```
plot(cug_trans_15)
```

Exceptionally strong friend-of-a-friend tendencies!



Betweenness centralization



R code (centralization)

R code (model)

Output

Plot output

```
### Conditioning on the dyad census  
snafun::g_centralize(fifa2006, measure = "betweenness")
```

```
$centralization  
[1] 0.1144484
```

```
$theoretical_max  
[1] 19421649
```

Betweenness centralization



R code (centralization)

R code (model)

Output

Plot output

```
### Conditioning on the dyad census
betw_f <- function(x, directed = FALSE) { # note: directed = FALSE!
  x <- snafun::fix_cug_input(x, directed = directed)
  snafun::g_centralize(x, measure = "betweenness", directed = directed)$centralization
}

cug_betw_06_ds <- sna::cug.test(fifa2006,
                               mode = "graph", FUN = betw_f,
                               cmode = "dyad.census", reps = 200)
```

OR

```
### Conditioning on the dyad census
cent <- function(x) snafun::g_centralize(x, measure = "betweenness")$centralization
cug_betw_06_ds <- snafun::stat_cug(snafun::to_igraph(fifa2006), FUN = cent,
                                   mode = "graph", cmode = "edges", reps = 200)
```

Betweenness centralization



R code (centralization)

R code (model)

Output

Plot output

Univariate Conditional Uniform Graph Test

Conditioning Method: `dyad.census`

Graph Type: `graph`

Diagonal Used: `FALSE`

Replications: `200`

Observed Value: `0.1144484`

$\Pr(X \geq \text{Obs})$: `0`

$\Pr(X \leq \text{Obs})$: `1`

Betweenness centralization

R code (centralization)

R code (model)

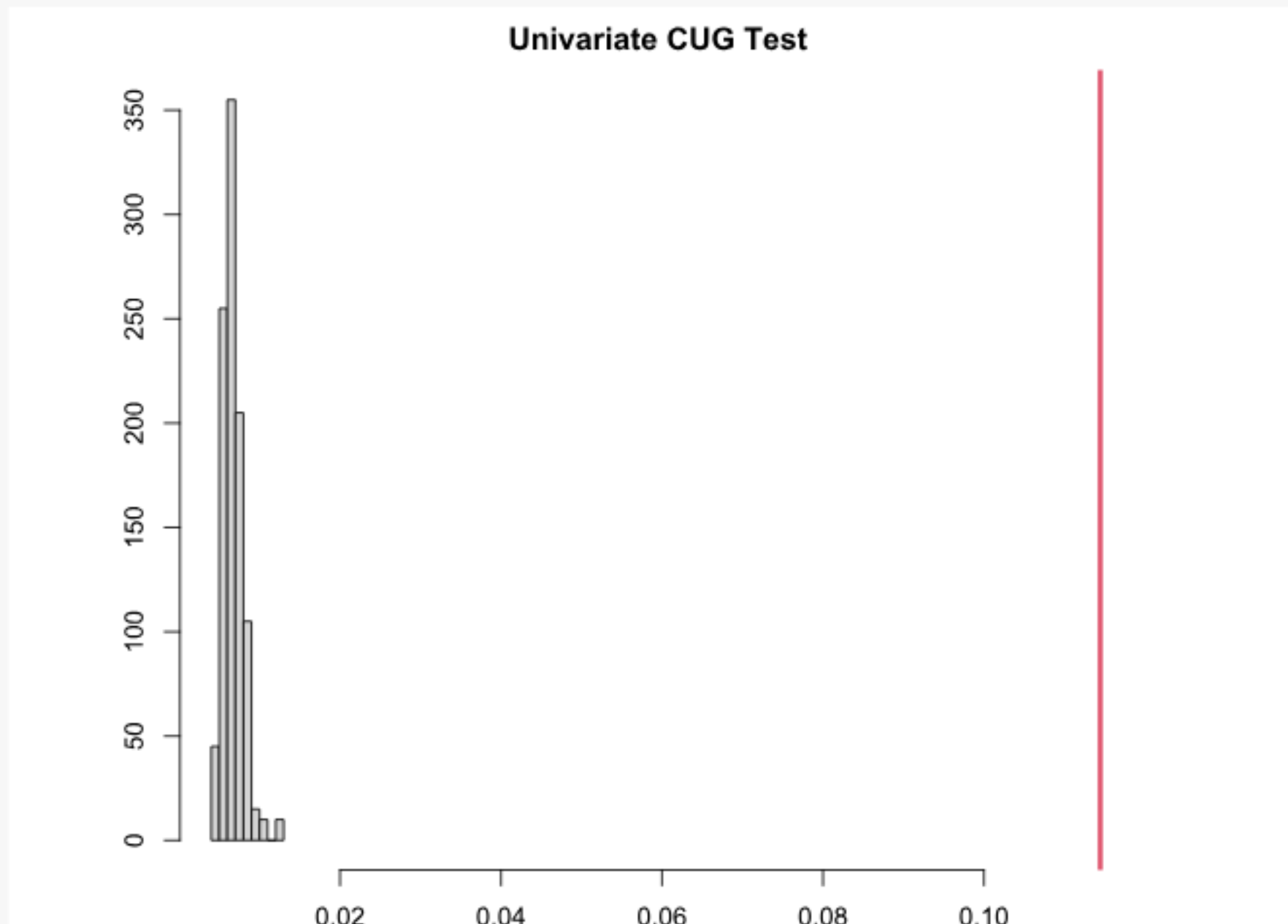
Output

Plot output

```
plot(cug_betw_06_ds)
```

We see exceptionally strong betweenness centralization in FIFA.

What does this imply about how FIFA operates?



Today's menu



Homeplay

CUG challenge

CUG challenge



Test if the number of walktrap communities in the fifa2015 network is statistically significant.

Recall:

```
data(fifa2015, package = "SNA4DSData")
betw_f <- function(x, directed = FALSE) { # note: directed = FALSE!
  x <- snafun::fix_cug_input(x, directed = directed)
  snafun::g_centralize(x, measure = "betweenness", directed = directed)$centralization
}

cug_betw_15_ds <- sna::cug.test(fifa2015,
                                mode = "graph", FUN = betw_f,
                                cmode = "dyad.census", reps = 200)
```

06:00

R code

Output

Plot output

```
walktrap_num_f <- function(x, directed = FALSE) { # note: directed = FALSE!
  x <- snafun::fix_cug_input(x, directed = directed)
  snafun::extract_comm_walktrap(x) |> length()
}

fifa_coms <- sna::cug.test(fifa2015, FUN = walktrap_num_f, mode = "graph",
                           diag = FALSE, cmode = "edges", reps = 200)

fifa_coms
plot(fifa_coms)
```

R code

Output

Plot output

```
print(fifa_coms)
```

Univariate Conditional Uniform Graph Test

Conditioning Method: edges

Graph Type: graph

Diagonal Used: FALSE

Replications: 1000

Observed Value: 25

Pr($X \geq \text{Obs}$): 0

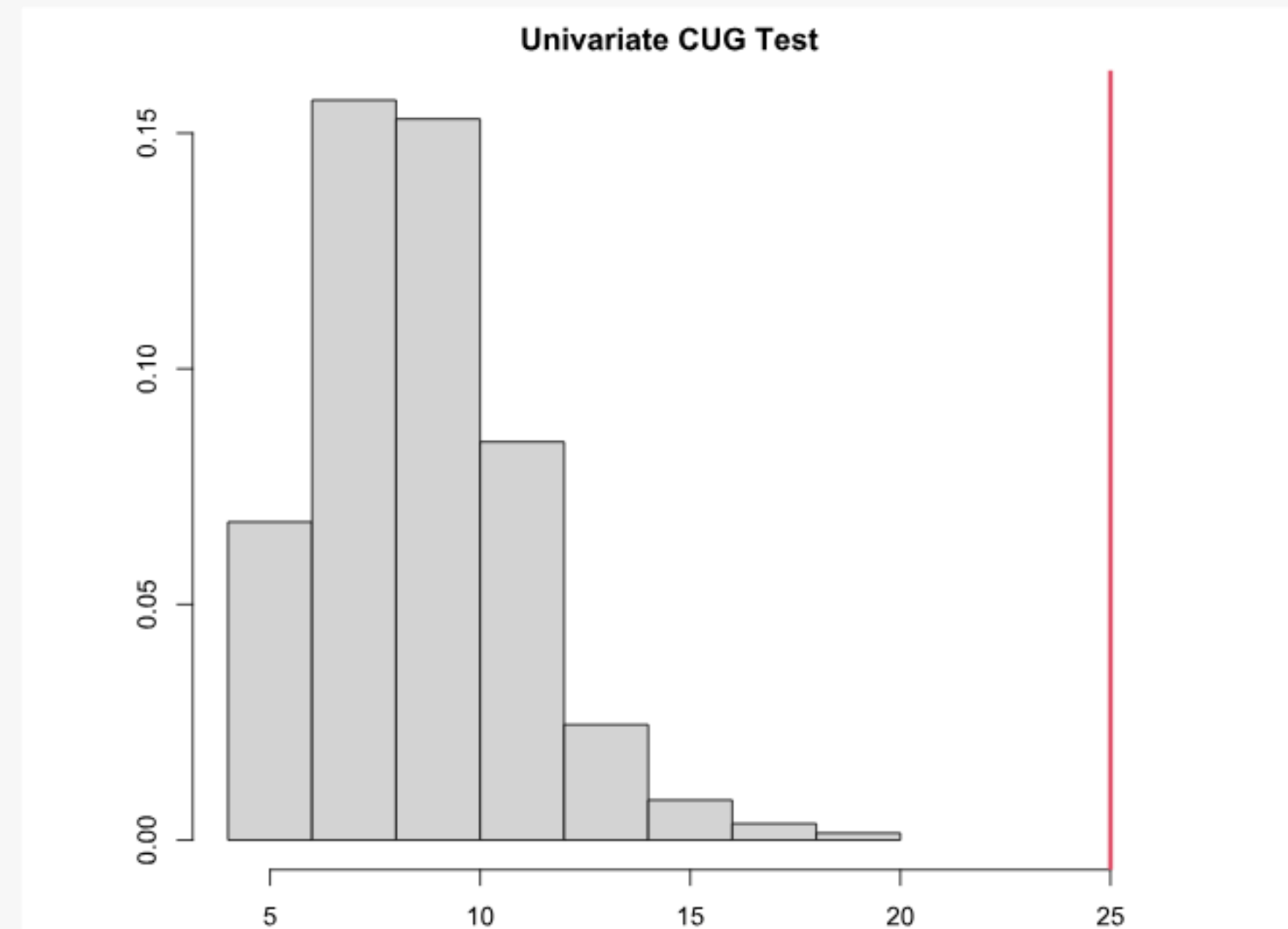
Pr($X \leq \text{Obs}$): 1

➔ There are more information-trapping communities (statistically significant) than you would expect of a network of this size and density.

R code

Output

Plot output



Today's menu



Homeplay

CUG challenge

Overview

Let's see if you have this mastered



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code: 4722 2476



Statistically modeling networks

Testing hypotheses

- use the simplest statistical model that suits your hypothesis test
 - QAP, MRQAP, Logistic QAP, and CUG are often great for simple hypotheses
 - use ERGM/GERGM/BTERGM to test more complex expectations

Estimating parameters

- MRQAP, Logistic QAP, and ERGM/GERGM/BTERGM allow for parameter estimation

Uncovering what the drivers are of the observed network

- ERGMs/GERGMs/BTERGMs model the network as a whole and allow for much more modeling flexibility and options
 - this comes at the challenge of fitting a more complex model